

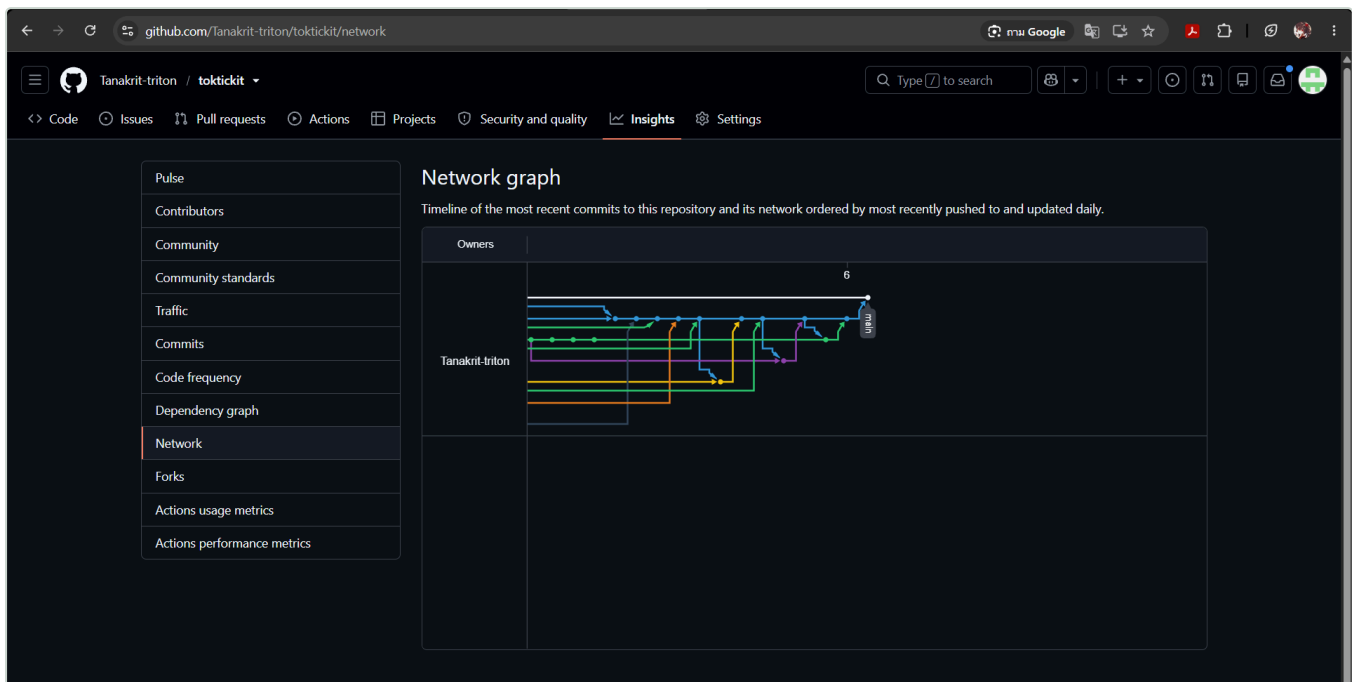
# TokTickIT — Lab 2 Submission

**Project:** TokTickIT — Requester Ticketing MVP with UI Foundation **Sprint:** Lab 2 **Author:** Tanakrit (67070503464) **Reviewer:** Richyboy170 **Repository:** <https://github.com/Tanakrit-triton/toktickit> **Released to** `main` : `#29`, merged as `09ee647`

## Answer Part 1

**Repository:** <https://github.com/Tanakrit-triton/toktickit>

**Commit history on final** `main`



`artifacts/lab-02/screenshots/process/desktop-branch-network.png`

GitHub's network graph, showing each feature branch diverging from and merging back into the integration line, and that line merging once into `main`.

|                                                                                   |          |         |
|-----------------------------------------------------------------------------------|----------|---------|
| Commits on Sep 6, 2026                                                            |          |         |
| Merge pull request #29 from Tanakrit-triton/lab2-staging                          | Verified | 09ee647 |
| Richyboy170 authored 8 hours ago                                                  |          |         |
| Merge pull request #28 from Tanakrit-triton/feat/lab-02-e2e-evidence              | Verified | 9ca2c6a |
| Richyboy170 authored 15 hours ago                                                 |          |         |
| Commits on Sep 5, 2026                                                            |          |         |
| Merge lab2-staging into feat/lab-02-e2e-evidence                                  |          | 787f89f |
| Tanakrit and claude committed yesterday                                           |          |         |
| Merge pull request #27 from Tanakrit-triton/feat/lab-02-ticket-detail-attachments | Verified | 4ba29b0 |
| Richyboy170 authored yesterday                                                    |          |         |
| Merge lab2-staging into feat/lab-02-ticket-detail-attachments                     |          | 2b9b20f |
| Tanakrit and claude committed yesterday                                           |          |         |
| Merge pull request #26 from Tanakrit-triton/feat/lab-02-my-tickets                | Verified | fb0ca5  |
| Richyboy170 authored 2 days ago                                                   |          |         |
| Merge pull request #25 from Tanakrit-triton/feat/lab-02-create-ticket             | Verified | a605024 |
| Richyboy170 authored 2 days ago                                                   |          |         |
| Merge lab2-staging into feat/lab-02-create-ticket                                 |          | 6d20e42 |
| Tanakrit and claude committed 2 days ago                                          |          |         |
| Merge pull request #24 from Tanakrit-triton/feat/lab-02-requester-context         | Verified | 9d16f8c |
| Richyboy170 authored 2 days ago                                                   |          |         |

artifacts/lab-02/screenshots/process/desktop-main-history-upper.png

|                                                                           |          |         |
|---------------------------------------------------------------------------|----------|---------|
| Merge pull request #25 from Tanakrit-triton/feat/lab-02-create-ticket     | Verified | a605024 |
| Richyboy170 authored 2 days ago                                           |          |         |
| Merge lab2-staging into feat/lab-02-create-ticket                         |          | 6d20e42 |
| Tanakrit and claude committed 2 days ago                                  |          |         |
| Merge pull request #24 from Tanakrit-triton/feat/lab-02-requester-context | Verified | 9d16f8c |
| Richyboy170 authored 2 days ago                                           |          |         |
| Merge pull request #23 from Tanakrit-triton/feat/lab-02-reference-api     | Verified | cbad261 |
| Richyboy170 authored 2 days ago                                           |          |         |
| Merge pull request #22 from Tanakrit-triton/feat/lab-02-data-model        | Verified | 2b4d2b9 |
| Richyboy170 authored 2 days ago                                           |          |         |
| Merge pull request #21 from Tanakrit-triton/chore/lab-02-test-tooling     | Verified | 08e5c20 |
| Richyboy170 authored 2 days ago                                           |          |         |
| Merge pull request #11 from Tanakrit-triton/docs/lab-02-spec              | Verified | 48ae4c7 |
| Richyboy170 authored 2 days ago                                           |          |         |
| docs(lab-02): add delivery evidence for the sprint                        |          | 3f3a606 |
| Tanakrit and claude committed 2 days ago                                  |          |         |

artifacts/lab-02/screenshots/process/desktop-main-history-lower.png

The commit history of the final `main` branch, in two portions: the release merge #29 down through #24, and #25 down to the specification merge #11. The pattern is the same throughout — a feature branch merges into `lab2-staging`, and `lab2-staging` merges once into `main`.

**Every merge commit is authored by Richyboy170, and each is Verified.** That is direct proof of the first course rule: the reviewer merges, not the author. It is a stronger claim than an approval, because an approval can precede a self-merge, whereas the author of the merge commit is a fact recorded in the object itself and cannot be edited after the fact.

Verifiable without the screenshots:

```
git log origin/main --merges --pretty=format:"%h %an | %s" | grep "lab-02"
```

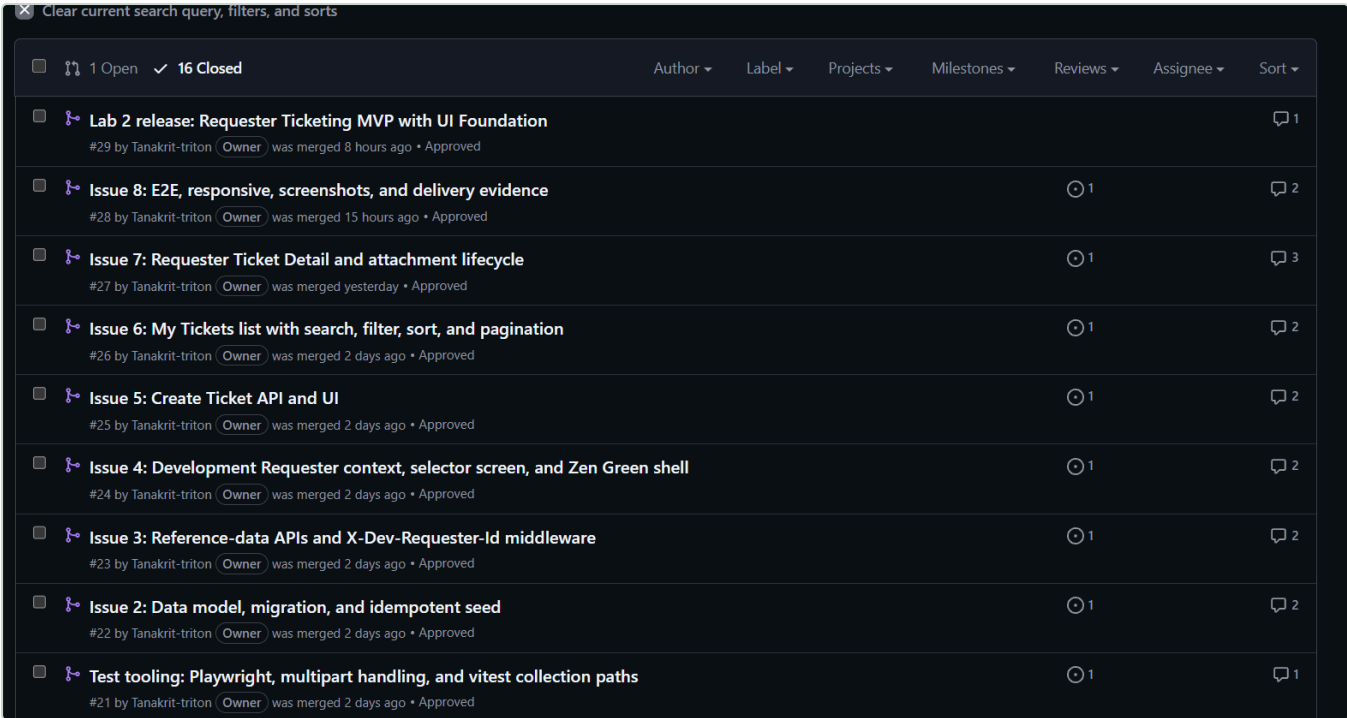
which returns, for all ten:

```
09ee647 Patiharn Liangkobkit | Merge pull request #29 from Tanakrit-triton/lab2-staging
9ca2c6a Patiharn Liangkobkit | Merge pull request #28 from Tanakrit-triton/feat/lab-02-e2e-evidence
4ba29b0 Patiharn Liangkobkit | Merge pull request #27 from Tanakrit-triton/feat/lab-02-ticket-detail-attachmen
fbd0ca5 Patiharn Liangkobkit | Merge pull request #26 from Tanakrit-triton/feat/lab-02-my-tickets
a605024 Patiharn Liangkobkit | Merge pull request #25 from Tanakrit-triton/feat/lab-02-create-ticket
9d16f8c Patiharn Liangkobkit | Merge pull request #24 from Tanakrit-triton/feat/lab-02-requester-context
cbad261 Patiharn Liangkobkit | Merge pull request #23 from Tanakrit-triton/feat/lab-02-reference-api
2b4d2b9 Patiharn Liangkobkit | Merge pull request #22 from Tanakrit-triton/feat/lab-02-data-model
08e5c20 Patiharn Liangkobkit | Merge pull request #21 from Tanakrit-triton/chore/lab-02-test-tooling
48ae4c7 Patiharn Liangkobkit | Merge pull request #11 from Tanakrit-triton/docs/lab-02-spec
```

Patiharn Liangkobkit is the account behind the handle Richyboy170. The Lab 1 release merge `99efd2d` is authored by Tanakrit-triton, which is the contrast: that sprint predates the rule.

| Merge                | Branch                                             | PR                  |
|----------------------|----------------------------------------------------|---------------------|
| <code>09ee647</code> | <code>lab2-staging</code> into <code>main</code>   | <a href="#">#29</a> |
| <code>9ca2c6a</code> | <code>feat/lab-02-e2e-evidence</code>              | <a href="#">#28</a> |
| <code>4ba29b0</code> | <code>feat/lab-02-ticket-detail-attachments</code> | <a href="#">#27</a> |
| <code>fbd0ca5</code> | <code>feat/lab-02-my-tickets</code>                | <a href="#">#26</a> |
| <code>a605024</code> | <code>feat/lab-02-create-ticket</code>             | <a href="#">#25</a> |
| <code>9d16f8c</code> | <code>feat/lab-02-requester-context</code>         | <a href="#">#24</a> |
| <code>cbad261</code> | <code>feat/lab-02-reference-api</code>             | <a href="#">#23</a> |
| <code>2b4d2b9</code> | <code>feat/lab-02-data-model</code>                | <a href="#">#22</a> |
| <code>08e5c20</code> | <code>chore/lab-02-test-tooling</code>             | <a href="#">#21</a> |
| <code>48ae4c7</code> | <code>docs/lab-02-spec</code>                      | <a href="#">#11</a> |

## Pull requests, all merged and approved



artifacts/lab-02/screenshots/process/desktop-pull-requests-merged.png

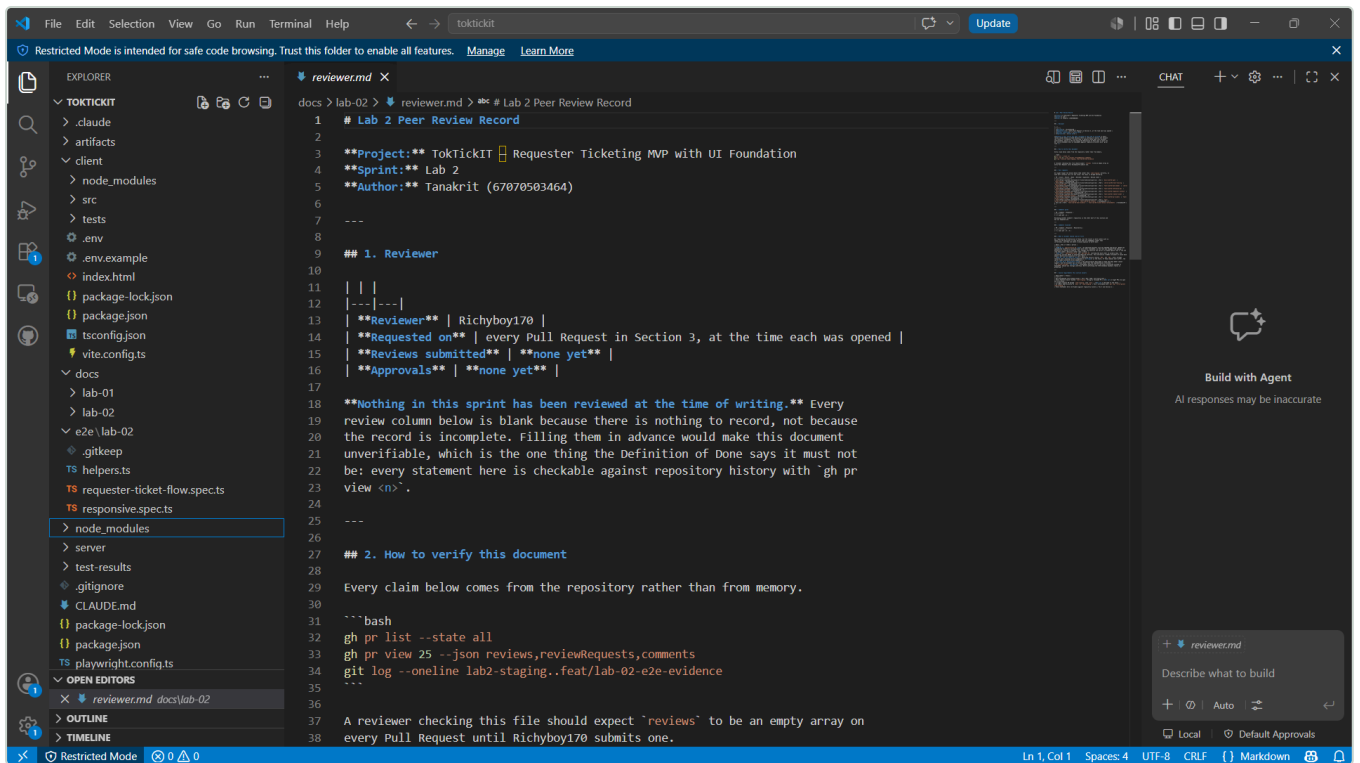
All ten Pull Requests, merged, each showing **Approved** and its linked Issue. This is supporting evidence for the same rule: the approvals are visible here, and the merge authorship above shows who acted on them.

## Issues and the Kanban board

Nine issues, one branch and one Pull Request each.

| Issue               | Title                                                                 |
|---------------------|-----------------------------------------------------------------------|
| <a href="#">#12</a> | Lab 2 sprint specification and test plan                              |
| <a href="#">#20</a> | Test tooling: Playwright, multipart handling, vitest collection paths |
| <a href="#">#13</a> | Data model, migration, and idempotent seed                            |
| <a href="#">#14</a> | Reference-data APIs and the X-Dev-Requester-Id middleware             |
| <a href="#">#15</a> | Development Requester context, selector screen, Zen Green shell       |
| <a href="#">#16</a> | Create Ticket API and UI                                              |
| <a href="#">#17</a> | My Tickets list with search, filter, sort, pagination                 |
| <a href="#">#18</a> | Requester Ticket Detail and attachment lifecycle                      |
| <a href="#">#19</a> | E2E, responsive, screenshots, and delivery evidence                   |

**Board:** <https://github.com/users/Tanakrit-triton/projects/2>

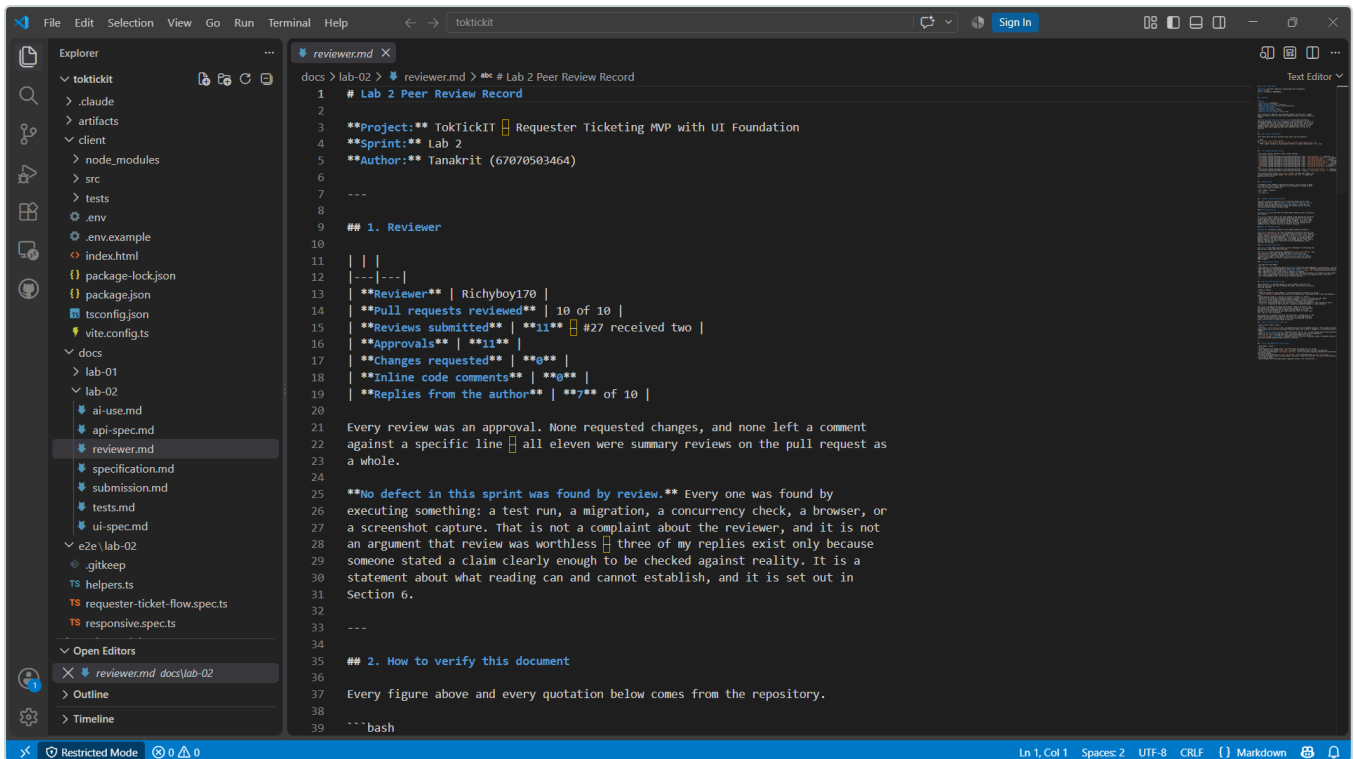


artifacts/lab-02/screenshots/project-board/desktop-kanban-done.png

**Done 13** — the nine Lab 2 cards and the four carried from Lab 1. Started, PR Review and Fixing are all empty, so no work is left in flight. Each card shows its issue number and the Pull Request that delivered it: #16 with #25, #17 with #26, #18 with #27, #19 with #28.

Issue state and board position are separate things in GitHub Projects: closing an issue does not move its card, and a card in Done does not close its issue. Both are true here, and the screenshot is the evidence for the second, since the board column is not derivable from the issue list.

## Repository structure



artifacts/lab-02/screenshots/process/desktop-repository-structure.png

The working tree: `client/` and `server/` hold the application, `docs/lab-02/` the four specification documents plus this submission, `e2e/lab-02/` the Playwright suites, `artifacts/` the captured evidence, and `CLAUDE.md` the course workflow rules that applied to every session. `node_modules/` and `test-results/` appear in the editor because they exist on disk; both are ignored and neither is tracked.

## Peer review record

**Rendered:** <docs/lab-02/reviewer.md>

Ten Pull Requests reviewed, **eleven reviews** — #27 received two — every one an approval, with no changes requested and no inline comments. Seven drew a reply from me. **Three of those replies corrected the review:** the purity claim on #22, the transaction claim on #25, and the mobile-cards claim on #26. Those exchanges are section 5 of that document and are the substantive part of the record.

Section 6 records something the sprint is worth being honest about: **no defect was found by review**. Every one was found by executing something — a test run, a migration, a concurrency check, a browser, a screenshot capture.

## This submission is itself awaiting review

The document you are reading, together with the corrected `reviewer.md` and the trimmed `ai-use.md`, lives on `docs/lab-02-submission` and is open as a Pull Request awaiting Richyboy170. It is not on `main` yet, and it is not merged.

Every link in this document therefore points at `blob/docs/lab-02-submission/` rather than `blob/main/`, so a marker following one today sees the corrected document rather than the superseded copy still on `main`.

**This is deliberate, and it is the same rule the rest of Part 1 is evidence for.** The submission could be merged by its author in a moment. Doing so would contradict the claim made three sections above — that every merge commit on `main` is authored by the reviewer — using the submission document itself as the counter-example. A record that documents a rule and then breaks it to publish itself is worth less than one that waits.

**Pull Request:** [#32](#)

One other Pull Request is open and unmerged: [#31](#), which fixes the end-to-end base URL default described in Part 3.

## README and .gitignore

**Rendered:** [README.md](#) and [.gitignore](#)

The README setup, run, seed and test instructions were executed from a clean checkout, not written from memory. `.gitignore` keeps attachment binaries and generated Playwright output out of the repository while keeping the screenshots under `artifacts/lab-02/screenshots/` tracked, because Part 1 of the Definition of Done requires them as evidence.

---

## Answer Part 2

Rendered specification: [docs/lab-02/specification.md](https://github.com/Tanakrit-triton/toktickit/blob/main/docs/lab-02/specification.md)

The document is numbered throughout, so each item can be cited from a test or a commit message:

| Section                     | Contents                                          | Count   |
|-----------------------------|---------------------------------------------------|---------|
| \$4 Functional Requirements | FR-01 ... FR-35                                   | 35      |
| \$5 Business Rules          | BR-01 ... BR-49                                   | 49      |
| \$9 Acceptance Criteria     | AC-01 ... AC-44                                   | 44      |
| \$10 Definition of Done     | Part 1 product completion, Part 2 course delivery | 2 parts |

\$11 additionally records the two SDS deviations (DEV-01, DEV-02), the sprint decisions (DEC-01 ... DEC-08), the assumptions (A-01 ... A-06), and the one amendment to an approved rule (AMD-01).

## The specification existed before implementation

Platform

Solutions

Resources

Open Source

Enterprise

Pricing

Search

Sign in

Sign up

Tanakrit-triton / toktickit

Public

Notifications

Fork 0

Star 0

<> Code

Issues

Pull requests

Actions

Projects

Security and quality

Insights

Commits

History for toktickit / docs / lab-02 / specification.md on main

All users

All time

Commits on Sep 4, 2026

feat(lab-02): add Ticket Detail screen and retire the A-06 constraints

Tanakrit and claude committed 2 days ago

3b852e7

<>

docs(lab-02): amend BR-05 to state the property, not the mechanism

Tanakrit and claude committed 2 days ago

9debaec

<>

feat(lab-02): allocate ticket numbers atomically in the database

Tanakrit and claude committed 2 days ago

6a5f015

<>

fix(lab-02): close the AC-17 gap, own the STY ids, stop attachments vanishing

Tanakrit and claude committed 2 days ago

2c0d9e7

<>

feat(lab-02): mount the shell in the application and add routing

Tanakrit and claude committed 3 days ago

2c152d6

<>

Commits on Sep 3, 2026

docs(lab-02): resolve identifier conflict and spec defects

Tanakrit and claude committed 3 days ago

9446f8b

<>

Commits on Sep 1, 2026

Update specification.md

artifacts/lab-02/screenshots/process/desktop-spec-precedes-implementation.png

The file history of `specification.md` on `main`, which carries absolute dates rather than the relative ones the pull request list shows. The earliest commit is **1 September 2026**. The implementation pull requests merged from **5 September** onwards:



| Pull request                         | Merged               |
|--------------------------------------|----------------------|
| <a href="#">#11</a> specification    | 2026-09-05T00:07:24Z |
| <a href="#">#22</a> data model       | 2026-09-05T00:13:08Z |
| <a href="#">#25</a> Create Ticket    | 2026-09-05T04:24:55Z |
| <a href="#">#27</a> Ticket Detail    | 2026-09-05T15:42:17Z |
| <a href="#">#28</a> E2E and evidence | 2026-09-06T05:52:08Z |

The specification was authored, reviewed and merged before any implementation pull request completed. Every document carried the status line *"Draft for approval — must be merged before implementation PRs begin"* until #11 merged.

**One honest qualification.** The specification did not survive contact with the code unchanged. Three rules in it proved unsatisfiable as written and were corrected during the sprint — BR-05 (amended as AMD-01), DEC-04, and the Unavailable attachment state in `ui-spec.md` §5.5. Each was found by attempting the implementation rather than by review. This is set out in Part 4, because it is the most useful thing the sprint produced.

---

## Answer Part 3

**Test results**, run on final `main`. The server and client commands are the documented README commands verbatim; the end-to-end command needed one override, for the reason set out below.

| Suite                | Command                                    | Tests      | Passed     | Failed   |
|----------------------|--------------------------------------------|------------|------------|----------|
| Unit + API           | <code>cd server &amp;&amp; npm test</code> | 92         | 92         | 0        |
| UI component + style | <code>cd client &amp;&amp; npm test</code> | 76         | 76         | 0        |
| Responsive + E2E     | <code>npm run test:e2e</code>              | 24         | 24         | 0        |
| <b>Total</b>         |                                            | <b>192</b> | <b>192</b> | <b>0</b> |

**These figures match** `tests.md` **section 6 exactly**. Nothing was adjusted to make them agree.

Captured output: [server-tests.txt](#), [client-tests.txt](#), [e2e-tests.txt](#)

**18 Playwright skips, none of them a disabled test**. Each is a spec declining to run at a viewport where it does not apply: E2E journeys run at one width (10), RSP-06 applies to mobile only (2), and `ui-spec.md` section 10 lists three screenshot paths at desktop only (6).

### One defect found while running these

`npm run test:e2e` **as documented fails on a clean setup**. The README says the UI runs on `http://localhost:5173`, and `playwright.config.ts` defaults its `baseURL` to 5173 — but `e2e/lab-02/helpers.ts` defaults `BASE` to **5174**:

```
e2e/lab-02/helpers.ts:6  export const BASE = process.env.E2E_BASE_URL ?? "http://localhost:5174";
playwright.config.ts:20  baseURL: process.env.E2E_BASE_URL ?? "http://localhost:5173",
```

Run against a UI on the documented port, every specification fails with `net::ERR_CONNECTION_REFUSED` at `http://localhost:5174`. The 5174 default was baked in during development because that is where Vite happened to be listening at the time, and the mismatch was never exercised because every later run passed `E2E_BASE_URL` explicitly.

The figures above were obtained with `E2E_BASE_URL=http://localhost:5173 npm run test:e2e`, which the README does document as the override for a UI on a different port — but 5173 is the documented default, so no override should be needed.

**This is reported rather than corrected in place**, because it is a defect in code already merged to `main`, and correcting it belongs in its own change with its own review rather than inside the submission document.

**The fix is written and awaiting review**. It is [#31](#), open against `main` from `fix/e2e-base-url-port`, which defaults `BASE` to the documented 5173 so that `npm run test:e2e` works with no override. I do not merge my own pull requests, so it stays open until the reviewer merges it.

`tests.md` §6 is unchanged: its figures were correct and remain correct.

# Answer Part 4

Rendered AI use document: <docs/lab-02/ai-use.md>

## LLM used

**Claude Opus 5** (model id `claude-opus-5`), through **Claude Code** in the Claude desktop application, with tool access to the working tree, a shell, and the `gh` CLI. That access is why the prompt table records outcomes rather than suggestions: most prompts ended in a command actually run against the real database and its real output. No other model or AI service was used.

## Key prompts

`ai-use.md` §2 tabulates **ten** prompts, quoted from the session and abridged only where long. Each row gives the prompt, what it was for, and what came back. They span the opening audit, the specification defect fixes, the `428` correction, Issue #13's false premise, the blocked dependency on Issue #15, and Issues #16 through #19.

## My Reflection

`ai-use.md` §5. The substance, in short:

The most useful thing the model did was not writing code. It was reading four documents against each other and finding that two contradicted each other on primary keys — a conflict that had survived being written, committed, and pushed to an open pull request.

The pattern held: the model was most valuable asked to **check** something against a written contract, and least trustworthy asked to **assert** that something worked. Every defect it found was found by comparison — specification against specification, generated SQL against the real state of the table, a green test run against what parallel execution actually does.

It produced defects of the same kind in its own work. Three tests in #18 passed before the feature existed, because they asserted only that something was absent and an empty response satisfies that. Two tests in #16 passed or failed by machine load. A regex built by string concatenation lost its escape and would have accepted a malformed ticket number.

§4 records the finding I consider most important. **Three rules from documents I wrote, reviewed and approved before any code existed described things the system could not do:**

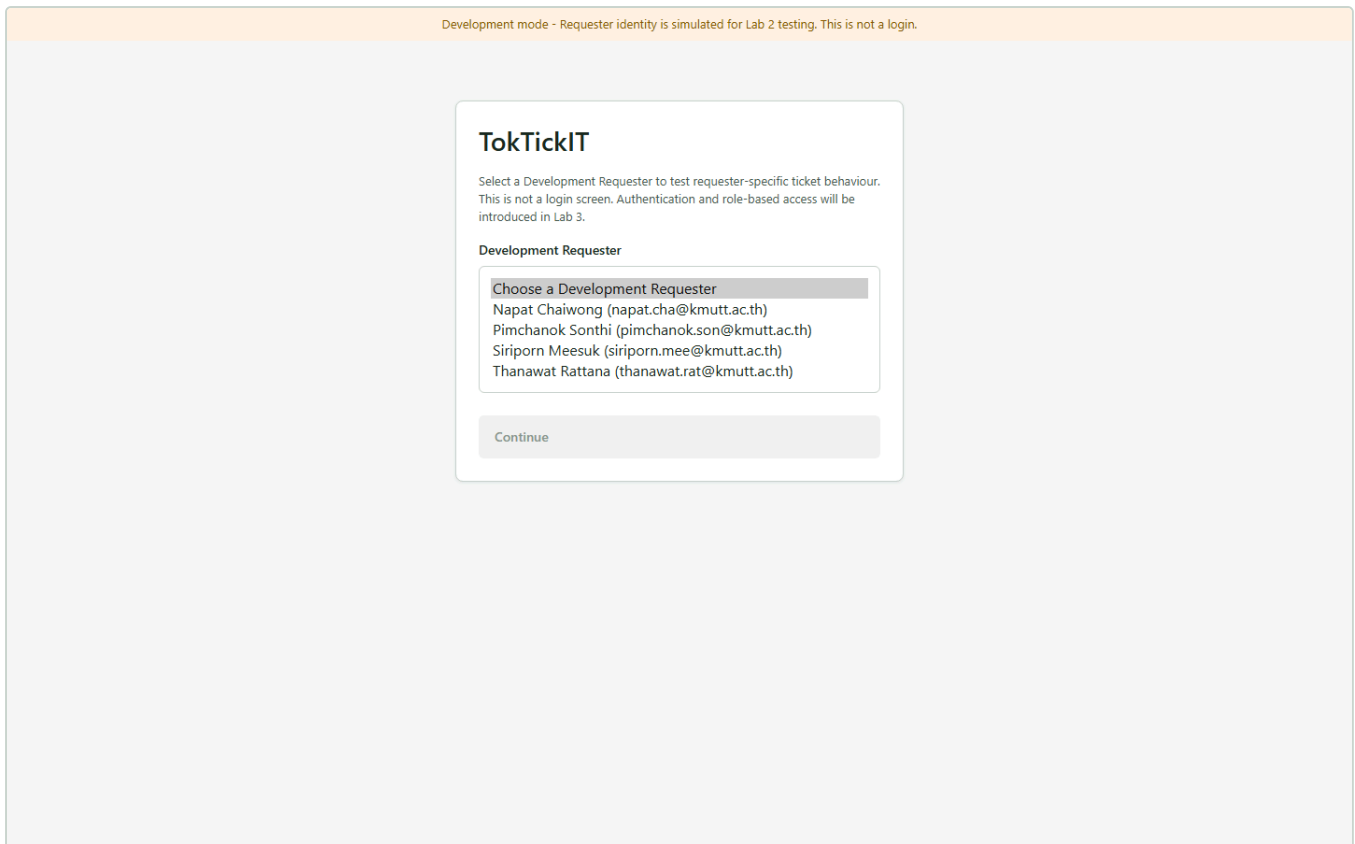
| Rule         | Why it could not hold                                                                                                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BR-05        | Implemented exactly, and duplicates were still reachable. Eight parallel creations failed six times. The rule named a mechanism and assumed a guarantee that mechanism does not deliver. Amended as AMD-01. |
| DEC-04       | Contradicted <code>api-spec.md</code> §1.2 on primary keys, on the same pull request, neither flagging it. Following it meant a destructive migration of Lab 1 data.                                        |
| ui-spec §5.5 | An attachment state shown "when an upload failed after the Ticket was created". A failed upload persists nothing, so a reloaded screen has nothing to render it from.                                       |

Each reads as correct, cites the right identifiers, and sounds like what an experienced engineer would write. Reviewing prose against prose cannot separate a rule that is correct from one that is merely plausible. The lesson I take is narrow: **a rule naming a mechanism is weaker than one stating a property**, because the mechanism can be implemented faithfully while the property it was chosen for goes unmet. BR-05 now states the property and names the test that proves it.

---

## Answer Part 5

**The simulated login screen — selecting the session Requester.** Zero points; graded within Part 6.



artifacts/lab-02/screenshots/requester-selection/desktop-loaded.png

The dropdown is expanded in this capture. A closed native select renders its popup through the operating system, so a screenshot of the closed control shows only the placeholder and proves nothing about which Requesters are listed. Only the control's presentation is changed; the options are the application's own, fetched from `GET /api/v1/dev-requesters`.

**Four active Requesters are listed** — Napat Chaiwong, Pimchanok Sonthi, Siriporn Meesuk, Thanawat Rattana. **The seeded inactive Requester, Kittipong Wong, is absent**, which is AC-01 and BR-10: inactive Requesters are filtered server-side and never reach the client at all.

**This is not authentication, and the screen says so.** The explanatory text reads "*This is not a login screen. Authentication and role-based access will be introduced in Lab 3.*", and the notice above it appears on every screen in the application. The selection is unsigned, held in session storage, and sent as the `X-Dev-Requester-Id` header; it carries no cryptographic guarantee and no part of the implementation treats it as proof of identity (BR-03, BR-11). DEC-02 keeps identity in a header rather than the URL or body, so the Lab 3 migration to a session cookie changes no route signature.

The remaining states of this screen — loading, empty, failure — and the selected-user display and Change Requester action are answered in Part 6.

## Answer Part 6

All captures are 1440x900 unless stated. Each was produced by a script that asserts its subject is painted and inside the viewport before the file is written, so a blank render or a subject out of frame cannot be produced.

Where the subject is the top of a screen the scroll position is reset before the shutter. Where the subject genuinely sits further down the page - the validation messages, the attachment rows - it is not, because resetting there would crop the subject out. The two loading captures are the one place where a skeleton is the subject rather than a thing asserted away.

### Q1 — the Requester field comes from the pre-entry selection, and the saved Ticket carries the matching requesterId

Siriporn Meesuk was chosen on the selector screen **before entering the application**. The Requester field on Create Ticket is filled from that choice and cannot be typed into:

TokTickIT My Tickets Create Ticket Acting as: Siriporn Meesuk [Change Requester](#)

Development mode - Requester identity is simulated for Lab 2 testing. This is not a login.

### Create Ticket

**Ticket information**

Ticket Number: Will be generated on submission

Ticket Date: September 7, 2026

Requester: Siriporn Meesuk

**Classification**

Category \*: Account and Access

Related System \*: Campus Wi-Fi

Requested Priority \*: High

**Problem detail**

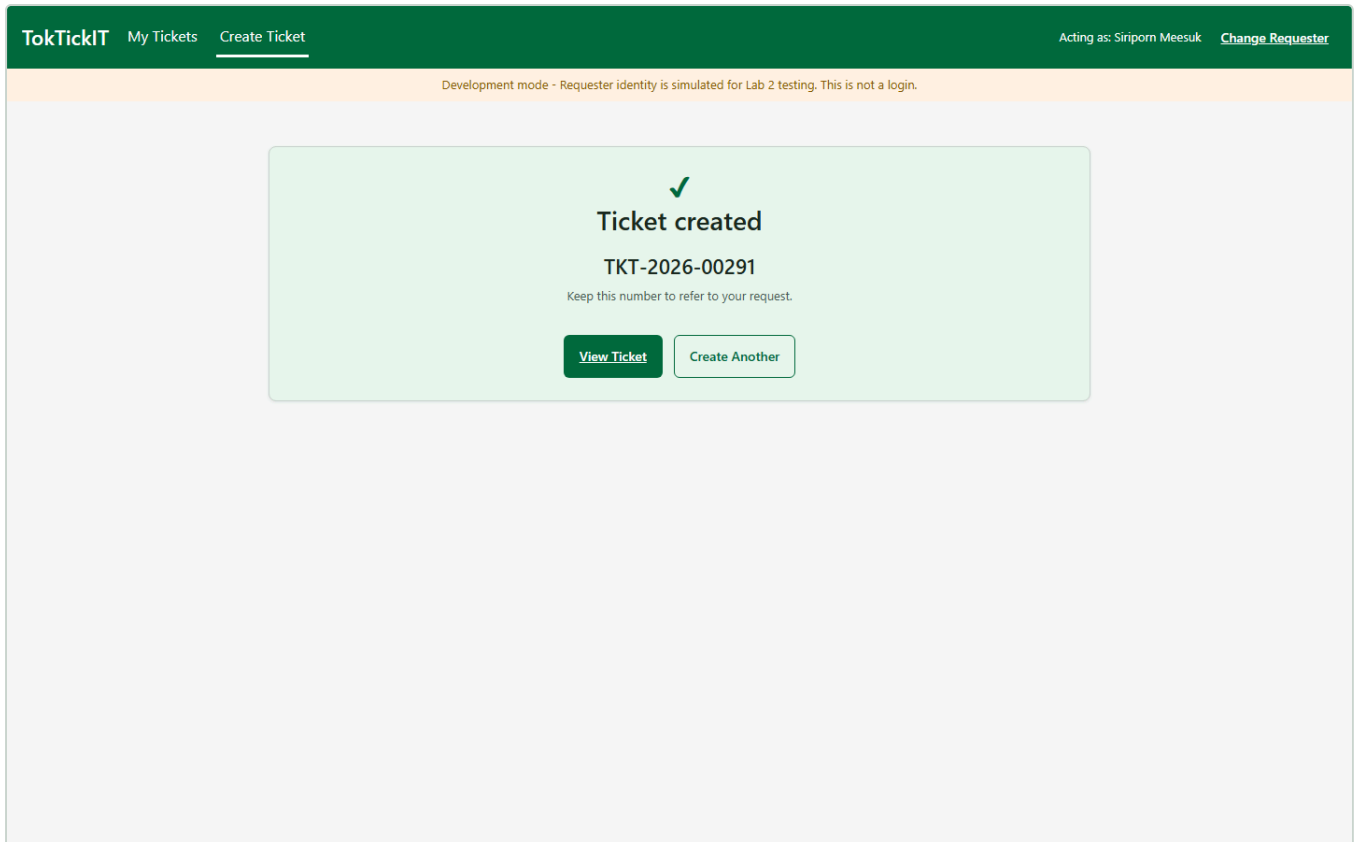
Ticket Summary \*: VPN drops every few minutes from the branch office

Description \*: Reported for the submission evidence with enough detail to satisfy the twenty character minimum.

artifacts/lab-02/screenshots/create-ticket/desktop-requester-field.png

The capture asserts the field holds Siriporn Meesuk **and** that its readOnly property is true before writing the file, so a writable field showing the right name could not have produced it.

Submitting the form returns the official number on the success panel:



artifacts/lab-02/screenshots/create-ticket/desktop-requester-ownership.png

The panel is from the run that produced `TKT-2026-00291`, which is the row quoted below. The form capture above is a separate run of the same flow, so the two are not one submission and are not offered as one.

The stored row, queried back out of PostgreSQL by that number ( `artifacts/lab-02/evidence/part-6-database-proof.txt` ):

```
ticketNumber      TKT-2026-00291
requesterId       30147b1b-4294-4293-988b-ecb614b31759
requester         Siriporn Meesuk
currentStatus     NEW
createdAt         2026-09-06T19:35:31.589Z
```

```
requesterId matches the selected Requester: true
status is NEW on creation (BR-02):             true
Ticket Number matches TKT-YYYY-NNNNN:         true
```

**Neither value is chosen by the client.** No `requesterId` and no `ticketNumber` appears in the create request body. Ownership is taken from the `X-Dev-Requester-Id` header on the server, and a client-supplied `requesterId` is ignored outright rather than consistency-checked, because checking it would invite clients to send it (BR-08). The number is allocated by the backend inside the same transaction as the insert, using an atomic database increment (BR-05, AMD-01). API-02 asserts the stored `requesterId` matches the header, API-06 asserts a body value is ignored, and API-41 proves eight concurrent creations all receive distinct numbers.

## Q2 — Create Ticket at desktop viewport, reference data loaded from the database

artifacts/lab-02/screenshots/create-ticket/desktop-initial.png

The system-generated group is populated and read-only: Ticket Number reads *"Will be generated on submission"*, Ticket Date is today, and Requester is the selected identity. Category and Related System come from `GET /api/v1/categories` and `GET /api/v1/related-systems`, both of which return active rows only, sorted by name.

**No option is hard-coded.** UI-10 asserts the rendered option list matches the API response **exactly, count included**, so a component carrying a fallback array alongside the API call fails rather than passing on a superset. API-38 and API-39 assert the endpoints return active rows only and that an inactive record never appears.



### Q3 — an invalid submission showing field-level messages

The screenshot shows a 'Create Ticket' form with the following fields and messages:

- Category \***: A dropdown menu with the text 'Select a Category'. Below it, the message 'Category is required.' is displayed.
- Related System \***: A dropdown menu with the text 'Select a Related System'. Below it, the message 'Related System is required.' is displayed.
- Requested Priority \***: A dropdown menu with the text 'Select a Requested Priority'. Below it, the message 'Requested Priority is required.' is displayed.
- Problem detail**: A section header.
- Ticket Summary \***: A text input field. Below it, the message 'Ticket Summary is required.' is displayed.
- Description \***: A large text area. Below it, the message 'Description is required.' is displayed.
- Attachments**: A section header.
- Attach evidence (optional)**: A dashed box containing the text 'JPG, PNG, WEBP, or PDF. Maximum 5 MB each, up to 5 files.' and a 'Choose Files' button. Below the button, the text 'No file chosen' is displayed.

artifacts/lab-02/screenshots/create-ticket/desktop-validation-failure.png

Submitting the empty form reports **all five failing fields at once**, each message directly below the control that failed, each field tinted with the invalid state. No request is sent (AC-12).

Placement is asserted, not just presence: STY-02 requires the message to be the **next sibling** of its own field and the field's `aria-describedby` target, so a single summary block at the top of the form — which `ui-spec.md` §2.1 forbids — would fail. The capture also asserts `document.activeElement` is the first failing field, so it cannot be produced unless focus really moved there.

The server enforces the same rules independently and returns every failing field in one `422` (BR-26, API-05). Client validation exists to make the feedback immediate; it never replaces the server, and a `422` replaces the client's messages with the server's per field (BR-24).

## Q4 — one valid and one invalid attachment selected, with the result explained

The screenshot shows a web form for creating a ticket. At the top is a dropdown menu labeled "Select a Requested Priority". Below it is a section titled "Problem detail" containing a "Ticket Summary \*" text field and a "Description \*" text area. The "Attachments" section follows, with a dashed box containing the text "Attach evidence (optional)" and "JPG, PNG, WEBP, or PDF. Maximum 5 MB each, up to 5 files." Below this is a "Choose Files" button and the text "No file chosen". There are two rows of attachments. The first row shows "battery-report.png" with a size of "8 B" and a "Remove" button. The second row shows "payload.exe" with a size of "2 B" and the text "File type not permitted" in red, along with a "Remove" button. At the bottom right are "Submit Ticket" and "Cancel" buttons.

artifacts/lab-02/screenshots/create-ticket/desktop-invalid-attachment.png

Two files were selected in one action: `battery-report.png` and `payload.exe`.

**What happened, and why.** `battery-report.png` was **accepted**. Its extension `.png` and its declared MIME type `image/png` are both permitted and they agree, and it is under 5 MB, so it appears as an ordinary selected row with its size and a Remove control, and it will be uploaded when the ticket is submitted.

`payload.exe` was **rejected**, and the row states why: **"File type not permitted"**. Neither its extension nor its declared type `application/x-msdownload` is among the four permitted combinations — JPG/JPEG, PNG, WEBP, PDF (BR-30). **The rejected file remains visible rather than being discarded silently**, which is the behaviour `ui-spec.md` §5.3 requires: a file that vanished without explanation would leave the Requester believing evidence had been attached. It is excluded from submission and occupies no slot against the five-file limit.

Both halves of the type check must pass **and agree**. A `.pdf` announced as an image is rejected too (UT-10), because either half alone would let a mislabelled file through. Size is checked separately: over 5 MB is rejected with "File exceeds 5 MB" (UT-11, API-26).

Writing this test found a real defect. The file input originally carried `accept=".jpg, .jpeg, .png, .webp, .pdf"`, which made the browser hide impermissible files from the picker — so the "File type not permitted" state `ui-spec.md` §5.3 specifies was **unreachable**. The filter was removed so that validation is the single gate and the specified state can actually occur.

The server enforces the same policy independently, so a client that skipped the check is refused: 415 for an impermissible or disagreeing type, 413 over 5 MB, 409 at the sixth active attachment (API-25, API-26, API-

27).

## Q5 — backend stopped, showing the safe error state with form values preserved

The screenshot shows a web form for creating a ticket. At the top, there are three input fields: 'Ticket Number' with the value 'Will be generated on submission', 'Ticket Date' with the value 'September 4, 2026', and 'Requester' with the value 'Siriporn Meesuk'. Below these is the 'Classification' section, which includes a 'Category' dropdown set to 'Network', a 'Related System' dropdown set to 'Campus Wi-Fi', and a 'Requested Priority' dropdown set to 'High'. The 'Problem detail' section contains a 'Ticket Summary' text area with the text 'Wi-Fi drops every few minutes in the library' and a 'Description' text area with the text 'The campus Wi-Fi disconnects every few minutes on the third floor of the library, on both laptop and phone.' At the bottom, there is an 'Attachments' section with a note: 'Attachments are added from Ticket Detail once the ticket has been created.' A red error message box at the bottom of the form states: 'The ticket could not be created. Your details have been kept - try again.' The 'Submit' button is visible at the bottom right of the form.

artifacts/lab-02/screenshots/create-ticket/desktop-api-failure.png

The form was filled, the API was then stopped **by process id**, and the submission attempted. The callout reads *"The ticket could not be created. Your details have been kept - try again."*

**Every entered value is still present** — Category, Related System, Requested Priority, Ticket Summary and Description all survive the failure, and Submit is re-enabled so the Requester can retry without re-entry (BR-27, AC-16). The capture asserts each field's value individually before writing.

**Nothing internal leaks.** The capture greps the rendered text for `500`, `ECONNREFUSED`, `localhost:3000` and stack frames, and refuses to write if any appears (BR-28). The thrown error is discarded by the screen rather than rendered.

Stopping the API **by PID** matters: stopping the `npm` wrapper leaves the `tsx watch` child holding port 3000, and the form would then submit successfully against a healthy API while the test claimed to have proved a failure state. E2E-05 polls until the port refuses connections before asserting anything.

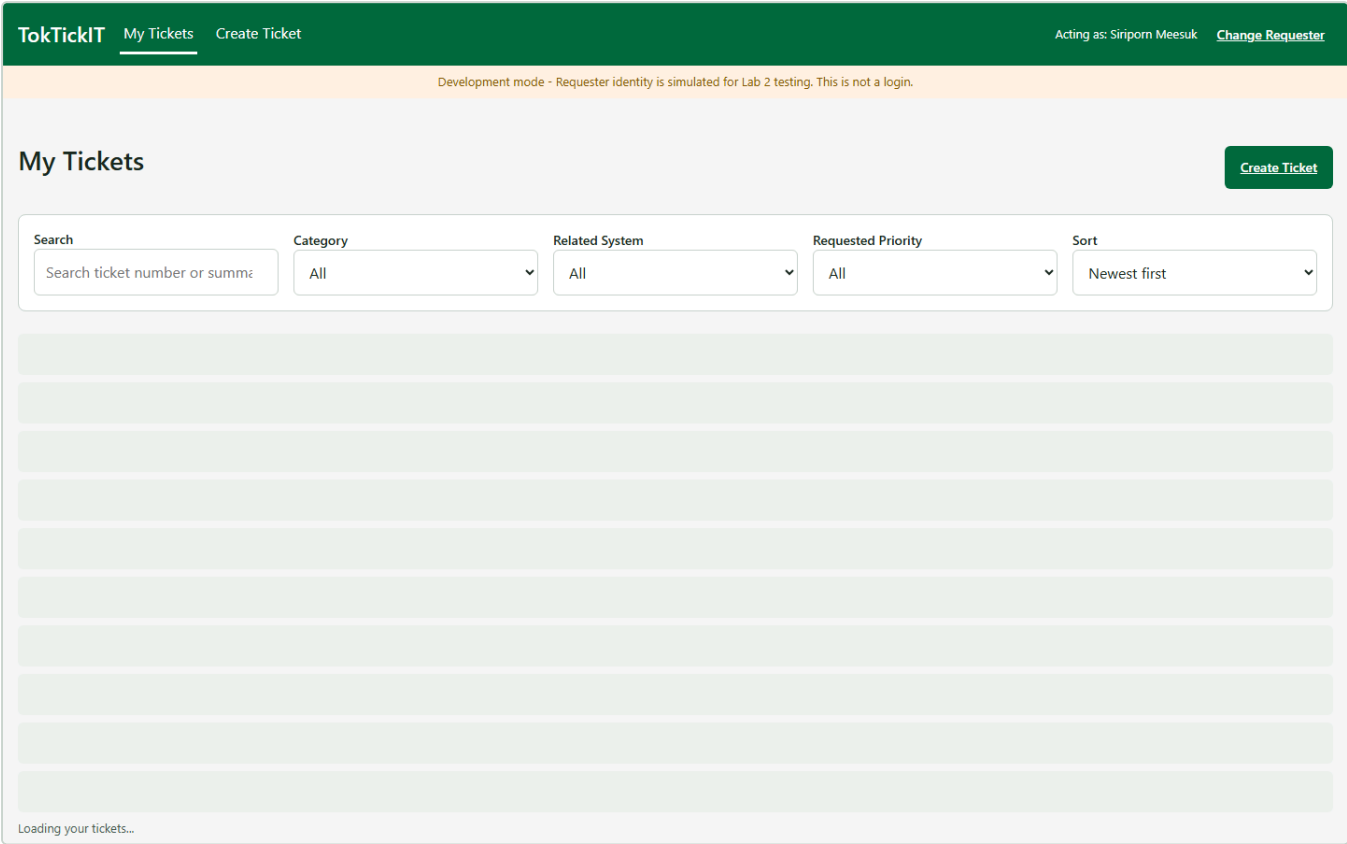
## The selector, the shell, and the remaining states

| Evidence                                                         | Screenshot                              |
|------------------------------------------------------------------|-----------------------------------------|
| Active-user dropdown, four active Requesters, inactive absent    | requester-selection/desktop-loaded.png  |
| Selected-user display and Change Requester in the shell          | requester-selection/desktop-shell.png   |
| Selector empty state — no active Requesters, no Continue offered | requester-selection/desktop-empty.png   |
| Selector failure state — safe message and Retry, no status code  | requester-selection/desktop-failure.png |
| Submitting state — Submit disabled, aria-busy , "Submitting..."  | create-ticket/desktop-submitting.png    |
| Success state — official Ticket Number and next actions          | create-ticket/desktop-success.png       |

The shell capture shows "Acting as: Siriporn Meesuk" with the Change Requester action beside it, and the development notice below the header. Changing Requester clears the selection and tears down the guarded subtree, so the previous Requester's data is discarded rather than reused (BR-12, UI-09).

## The loading state

The loading state exists only while a request is in flight, so at full speed a capture of it is a race rather than evidence. The network was therefore throttled through the DevTools protocol - 2.5 s of added latency at 15 kbit/s - **after** the application bundle had loaded, so only the data request is slowed and the skeleton stays on screen long enough to photograph.



artifacts/lab-02/screenshots/my-tickets/desktop-loading.png

My Tickets while the list is in flight: skeleton rows under `aria-busy="true"` and the message "*Loading your tickets...*". The capture asserts the ticket table is **absent**, so this cannot be a skeleton painted over a list that had already arrived.

artifacts/lab-02/screenshots/create-ticket/desktop-loading.png

Create Ticket while the reference data is in flight: Category and Related System are **disabled** and read "*Loading...*", so neither can be chosen before its options exist. Requested Priority sits enabled beside them because it is a fixed enumeration in the client and waits on no request - the difference is deliberate, not an inconsistency.

These captures show the state; they do not replace the tests that assert it. UI-02 and UI-26 continue to enforce the selector and list loading states in the counted suite, where they run at full speed and do not depend on throttling.

## Answer Part 7

---

### Requester-scoped ownership: one Requester cannot see another tickets.

A pair, so the disappearance is visible rather than asserted. Both are the same search for `TKT-2026-00290`; only the acting Requester differs.

| Evidence                                                        | Screenshot                                                                                    |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Requester A</b> (Siriporn Meesuk) sees the ticket            | <code>artifacts/lab-02/screenshots/ownership/desktop-requester-a-sees-ticket.png</code>       |
| <b>Requester B</b> (Napat Chaiwong), same search, finds nothing | <code>artifacts/lab-02/screenshots/ownership/desktop-requester-b-cannot-see-ticket.png</code> |
| Direct URL to another Requester ticket is refused               | <code>artifacts/lab-02/screenshots/ownership/desktop-direct-url-refused.png</code>            |

The refusal capture additionally asserts the ticket number does **not** appear anywhere on the page, so the refusal cannot leak what it is refusing.

Tested by **E2E-02** (switching Requester replaces the visible list) and **E2E-03** (typing a foreign ticket URL is refused), plus **API-10**, **API-21** and **API-22**.

---

## Answer Part 8

### Attachment lifecycle, and unauthorized attachment access.

| Evidence                                                  | Screenshot                                                                             |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------|
| Ticket Detail, read-only region with zero controls        | <code>artifacts/lab-02/screenshots/ticket-detail/desktop-view.png</code>               |
| Removed attachment: metadata and reason kept, no Download | <code>artifacts/lab-02/screenshots/ticket-detail/desktop-attachment-removed.png</code> |
| Removal modal with required reason                        | <code>artifacts/lab-02/screenshots/ticket-detail/desktop-removal-modal.png</code>      |
| Attachment unreachable to a non-owner                     | <code>artifacts/lab-02/screenshots/ownership/desktop-attachment-unreachable.png</code> |

The attachment refusal has no screen of its own, and that is a property of the design rather than a gap: the whole ticket is already unreachable, so the attachment list is never rendered. The API evidence behind the screenshot is captured as text in [ownership-refusals.txt](#):

|                                              |          |
|----------------------------------------------|----------|
| owner downloads their own attachment         | HTTP 200 |
| another Requester downloads the same         | HTTP 404 |
| another Requester requests the ticket        | HTTP 404 |
| the same Requester requests a missing ticket | HTTP 404 |

The last two bodies are identical, which is the point of DEC-01.

Soft removal is proved end to end by **E2E-04**: the row is retained with its reason and timestamp, the binary is deleted from disk, and a direct download of a removed attachment returns `410`.

## Answer Part 9

---

**Rendered UI specification:** <docs/lab-02/ui-spec.md>

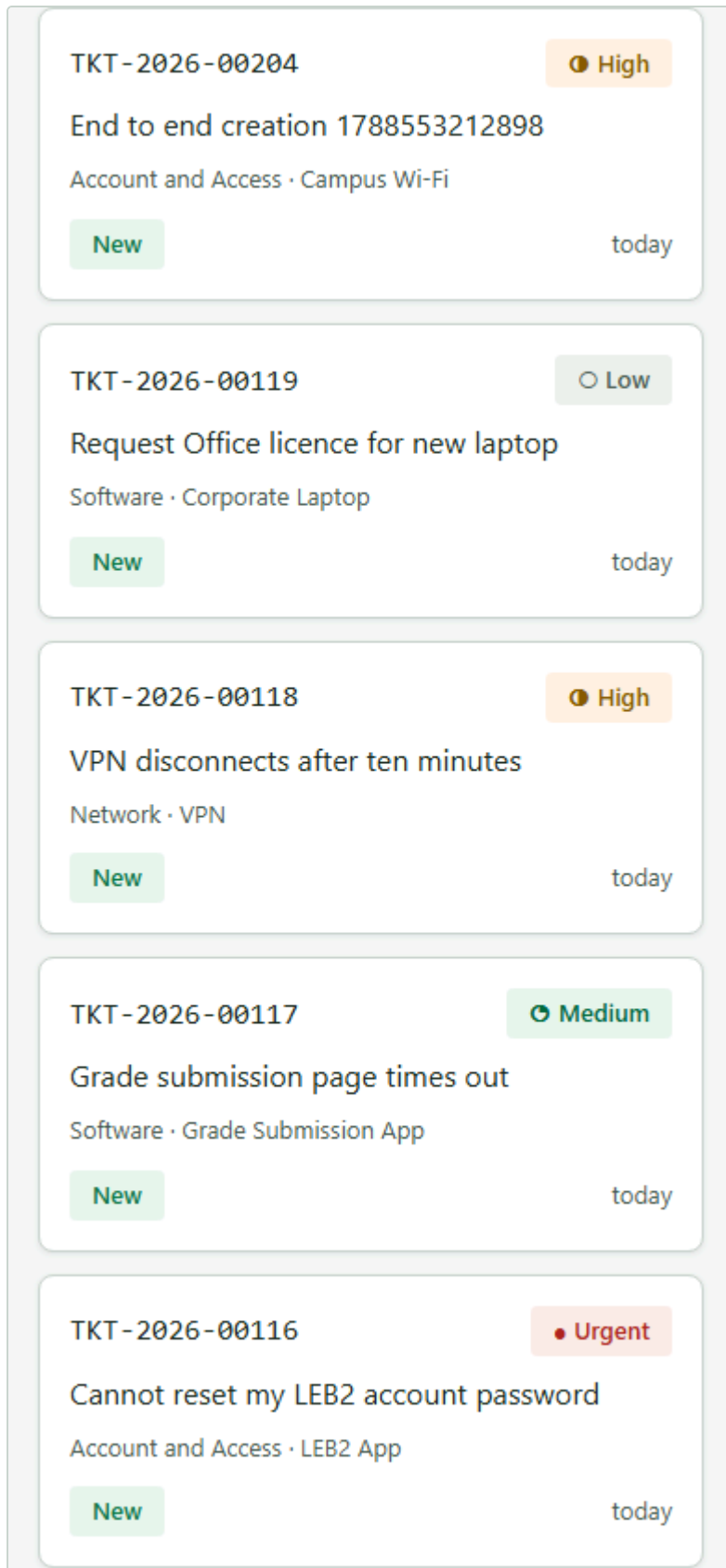
It fixes the Zen Green tokens (§1), the six control states (§2), the button hierarchy (§3), the shell (§4), every screen (§5), badges (§6), responsive rules (§7), accessibility (§8), test hooks (§9), and the screenshot paths (§10).

### The three viewports

Desktop 1440x900, tablet 834x1112, mobile 390x844 — the widths §10 fixes.

| Screen        | Desktop                                        | Tablet                                        | Mobile                                        |
|---------------|------------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| Create Ticket | <code>create-ticket/desktop-initial.png</code> | <code>create-ticket/tablet-initial.png</code> | <code>create-ticket/mobile-initial.png</code> |
| My Tickets    | <code>my-tickets/desktop-populated.png</code>  | <code>my-tickets/tablet-populated.png</code>  | <code>my-tickets/mobile-cards.png</code>      |
| Ticket Detail | <code>ticket-detail/desktop-view.png</code>    | <code>ticket-detail/tablet-view.png</code>    | <code>ticket-detail/mobile-view.png</code>    |





artifacts/lab-02/screenshots/my-tickets/mobile-cards.png

The mobile list renders **cards, not a table**. That is a difference in the DOM rather than in styling: AC-40 asks for cards "rather than a table", and a table hidden or scrolled by CSS is still a table in the accessibility tree. The capture asserts **zero** `table` elements anywhere in the document, every card at least 44px tall, and no horizontal page overflow. The tablet table drops Related System only; every other column is retained.

## Completed visual checklist

`tests.md` §4, all 22 rows at all three widths. Rows marked **auto** are enforced by a test that fails the build rather than by a human eye, and the test is named; rows marked n/a do not apply at that width and say why rather than being ticked vacuously.

| Requirement                          | How it is covered                                                                                                                                                                                                            |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Colours</b>                       | STY-11 parses the token table out of <code>ui-spec.md</code> itself and fails on any colour in the stylesheet absent from it. The 15 distinct values are all present, and none of the D-09 KMUTT values appears in any file. |
| <b>Editable and read-only fields</b> | STY-03. Read-only controls carry the <code>readonly</code> attribute and the read-only background; editable ones do not. Read-only stays legible rather than greyed into illegibility.                                       |
| <b>Disabled versus read-only</b>     | STY-04. Disabled cannot take focus; read-only can. Two states, not one greyed appearance.                                                                                                                                    |
| <b>Validation placement</b>          | STY-01 and STY-02. Every required field carries a red asterisk, and each message is the next sibling of its own field and its <code>aria-describedby</code> target.                                                          |
| <b>Button hierarchy</b>              | STY-06 asserts exactly one primary button per screen; STY-07 asserts no control relies on an icon alone.                                                                                                                     |
| <b>Clipping and overlap</b>          | RSP-01 ... RSP-03 at all three widths, plus the visual pass against the screenshots.                                                                                                                                         |
| <b>Horizontal overflow</b>           | RSP-01 ... RSP-03 assert <code>scrollWidth - clientWidth &lt;= 1</code> at every width.                                                                                                                                      |

**One deliberate exception.** `/lab-01` carries no development notice and uses Bootstrap colours outside the token table. A-05 excludes it from AC-44 and from this checklist: wrapping the retained Lab 1 page in the Zen Green shell would change the slice A-04 promises to preserve.

**Two limits worth stating.** `jsdom` loads no stylesheet, so the STY tests assert the attribute and class contract the stylesheet keys off, plus the rules' presence in source — what `jsdom` cannot see, that the rules paint, is covered by the responsive suite and these screenshots. And there are no pixel-diff baselines: they would fail on every intentional style change during a sprint that changed styles constantly (`tests.md` §7).